

Практикум

D6:archive

Twenty-nine questions, 114 marks. Every question the archive asks about a random variable given by a probability density function, from May 2021 to November 2025, in the order of the six techniques rather than the order of the papers.

О практикуме

Плотностная половина

statistics.continuous_random_variables: 90 баллов из 199 (0.452) — всё, где величина задана формулой, а не кривой с μ и σ . 10 баллов statistics.expected_value (0.167 — 10 из 60): ожидаемая сумма покупки и E , Var по плотности. 14 баллов statistics.probability (0.059 — 14 из 237): три условные под плотностью и доля времени в синусоидальной модели. 114 баллов, 35 блоков, и дубля нет — впервые за шесть практикумов статистики. 95 баллов на Paper 2, 19 на Paper 1; из 95 калькуляторных 29 — алгебра «show that», точное $E(X)$ и формулы через n .

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d6-density

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D6 archive — the probability density function, all of it

Twenty-nine questions, 114 marks.

Every question the archive asks about a random variable given by a probability density function, from May 2021 to November 2025, in the order of the six techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-d6-density.ipynb`.

§	technique	questions	marks
1	The area	5	13
2	The constant	5	24
3	The median, a quartile	7	27
4	The mode	3	10
5	The mean and the variance	6	30
6	A second question around an area	3	10

The checks are the same ones the practicum uses and they store nothing: each is handed the density exactly as the question prints it — `Density({interval: formula})`, zero elsewhere — and finds the area, any letter in the density, the median, the mode and the moments itself. A letter nobody gives a value is found from the area under f being one. Three significant figures, unless the question asks otherwise.

Parts of one question are split by technique. May 2024 TZ2 Q10 appears in §§ 1, 3, 4 and 5; each part repeats what it needs.

A *show that* is checked by its last line before the given result: the area under f as an expression in the letters — the thing you set equal to one — or $f(9)$ in terms of a and b .

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *            # checks +
Density, P(), Expect, Var, SD

language('en')               # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

§ 1. The area

Five questions, 13 marks. The density is known: an integral over the right interval.

1.1 — May 2023 TZ1 Paper 2 Q11(b), 2 marks

A continuous random variable, X , has a probability density function defined by

$$f(x) = \begin{cases} \frac{6}{\pi\sqrt{16-x^2}}, & 0 \leq x \leq 2 \\ 0, & \text{otherwise.} \end{cases}$$

(b) Find $P(X < 0.5)$.

1.2 — May 2024 TZ2 Paper 2 Q10(b), 2 marks

The weight, in kilograms, of chocolates bought by a random customer can be modelled by a continuous random variable X with probability density function

$$f(x) = \begin{cases} \frac{6}{85} (4 + 3x - x^2), & 0.5 \leq x \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

(b) Find $P(1 \leq X \leq 2)$.

```
q1_1 = ...          # P(X < 0.5)
q1_2 = ...          # P(1 <= X <= 2)

X = Density({(0, 2): 6/(pi*sqrt(16 - x**2))},
            'X')
W = Density({(Rational(1, 2), 3): Rational(6,
85)*(4 + 3*x - x**2)}}, 'X')

verify_chance('1.1', q1_1, P(X < 0.5))
verify_chance('1.2', q1_2, P((W >= 1) & (W <=
2)))
```

1.3 — May 2025 TZ1 Paper 2 Q11(b), 2 marks

In a marathon race, the random variable T represents the time, in hours, taken for a runner to complete the race. The probability density function for T is

$$f(t) = \begin{cases} \frac{4}{21} \left(1 - \cos \left(\frac{4\pi}{9}(t - 2.25) \right) \right), & 2.25 \leq t < 4.5 \\ \frac{4}{21} \left(1 + \cos \left(\frac{\pi}{3}(t - 4.5) \right) \right), & 4.5 \leq t \leq 7.5 \\ 0, & \text{otherwise.} \end{cases}$$

The runners who finish the race in 3.5 hours or less are considered to be fast runners.

(b) Find the probability that a runner chosen at random is a fast runner.

```
q1_3 = ...          # P(T <= 3.5)

T = Density({(Rational(9, 4), Rational(9,
2)): Rational(4, 21)*(1 - cos(4*pi/9*(t -
Rational(9, 4)))),
              (Rational(9, 2), Rational(15,
2)): Rational(4, 21)*(1 + cos(pi/3*(t -
Rational(9, 2))))},
            'T', var=t)

verify_chance('1.3', q1_3, P(T <= 3.5))
```

1.4 — May 2024 TZ2 Paper 2 Q10(d), 3 marks

The weight, in kilograms, of chocolates bought by a random customer has probability density function

$$f(x) = \begin{cases} \frac{6}{85} (4 + 3x - x^2), & 0.5 \leq x \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

The shop sells chocolates to customers at \$25 per kilogram. However, if the weight of chocolate bought by a customer is at least 0.75 kilograms, the shop sells chocolate at a discounted rate of \$24 per kilogram.

(d) Find the probability that a randomly selected customer spends at most \$48.

**1.5 — May 2023 TZ2 Paper 2 Q10(e),
4 marks**

The height, H metres, of the base of a weight on a spring above the ground is $H(t) = a \cos(7.8t) + b$, $0 \leq t \leq 10$, where t is the time in seconds after the weight is released. It is released at its minimum height of 1 metre and reaches a maximum height of 1.8 metres, so that $a = -0.4$ and $b = 1.4$. A camera takes a picture of the weight at a random time during the first five seconds of its motion.

(e) Find the probability that the height of the base of the weight is greater than 1.5 metres at the time the picture is taken.

```
q1_4 = ...          # P(spend <= 48)
q1_5 = ...          # P(H > 1.5)

W = Density({(Rational(1, 2), 3): Rational(6,
85)*(4 + 3*x - x**2)}, 'X')
spend = W.map(lambda w: Piecewise((25*w, w <
0.75), (24*w, True)), 'spend')
clock = Density({(0, 5): Rational(1, 5)},
'T', var=t)
H = clock.map(lambda s: -0.4*cos(7.8*s) +
1.4, 'H')

verify_chance('1.4', q1_4, P(spend <= 48))
verify_chance('1.5', q1_5, P(H > 1.5))
```

§ 2. The constant

Five questions, 24 marks. A letter in f : the whole area is one.

2.1 — May 2021 TZ1 Paper 2 Q7, 7 marks

A continuous random variable X has the probability density function

$$f(x) = \begin{cases} \frac{x}{\sqrt{(x^2 + k)^3}}, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

where $k \in \mathbb{R}^+$.

(a) Show that $\sqrt{16+k} - \sqrt{k} = \sqrt{k} \sqrt{16+k}$.

(b) Find the value of k .

For (a) enter the area under f in terms of k .

```

k = symbols('k', positive=True)

q2_1a = ...      # the area under f, in terms
of k
q2_1b = ...      # k

X = Density({(0, 4): x/sqrt((x**2 + k)**3)},
'X')

verify_chance('2.1(a)', q2_1a,
total_probability(X), free=k)
verify_letters('2.1(b)', q2_1b, k, [X])

```

2.2 — May 2022 TZ1 Paper 1 Q7(a), 4 marks

The continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{k}{\sqrt{4 - 3x^2}}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of k .

2.3 — November 2022 Paper 2 Q6(a), 5 marks

The continuous random variable X has a probability density function $f(x) = axe^x$ for $0 \leq x \leq b$, and 0 otherwise, where $a, b \in \mathbb{R}^+$.

(a) Find an expression for a in terms of b .

```
k = symbols('k', positive=True)
a, b = symbols('a b', positive=True)

q2_2 = ...          # k, exact
q2_3 = ...          # a in terms of b

K = Density({(0, 1): k/sqrt(4 - 3*x**2)},
            'X')
A = Density({(0, b): a*x*exp(x)}, 'X')

verify_letters('2.2', q2_2, k, [K],
exact=True)
verify_letters('2.3', q2_3, a, [A], free=b)
```

2.4 — May 2024 TZ1 Paper 2 Q8(a), 2 marks

A continuous random variable X has a probability density function

$$f(x) = \begin{cases} kx, & 0 \leq x \leq k \\ 2kx - x^2, & k < x \leq 2k \\ 0, & \text{otherwise} \end{cases}$$

where $k > 0$.

(a) Show that k satisfies the equation $7k^3 = 6$.

Enter the area under f in terms of k .

2.5 — November 2025 TZ3 Paper 2
Q11(a), (b)(i), 6 marks

The lengths, in metres, of jumps in a long jump competition can be modelled by a continuous random variable X with probability density function

$$f(x) = \begin{cases} a(x-3)^3 + b(x-3)^2, & 3 \leq x \leq 9 \\ 0, & \text{otherwise} \end{cases}$$

where $a, b \in \mathbb{R}$, $a, b \neq 0$.

(a) Show that $324a + 72b = 1$.

It is given that $f(9) = 0$.

(b) (i) Show that $6a + b = 0$.

For (a) enter the area under f in terms of a and b ; for (b)(i), $f(9)$ in terms of a and b .

```

k = symbols('k', positive=True)
a, b = symbols('a b')

q2_4 = ...          # the area under f, in terms
of k
q2_5a = ...         # the area under f, in terms
of a and b
q2_5b = ...         # f(9) in terms of a and b

B = Density(({(0, k): k*x, (k, 2*k): 2*k*x -
x**2}, 'X'))
J = Density(({(3, 9): a*(x - 3)**3 + b*(x -
3)**2}, 'X'))

verify_chance('2.4', q2_4,
total_probability(B), free=k)
verify_chance('2.5(a)', q2_5a,
total_probability(J), free=[a, b])
verify_identity('2.5(b)(i)', q2_5b, J.pdf(9))

```

§ 3. The median, a quartile

Seven questions, 27 marks. A boundary with a share of the area to its left.

3.1 — November 2021 Paper 2 Q7, 6 marks

A continuous random variable X has a probability density function $f(x) = \arccos x$ for $0 \leq x \leq 1$, and 0 otherwise. The median of this distribution is m .

(a) Determine the value of m .

(b) Given that $P(|X - m| \leq a) = 0.3$, determine the value of a .

```

q3_1a = ...      # m
q3_1b = ...      # a

m = symbols('m')
a = symbols('a', positive=True)
X = Density({(0, 1): acos(x)}, 'X')
median = Eq(P(X < m), 0.5)

verify_letters('3.1(a)', q3_1a, m, [X],
[median])
verify_letters('3.1(b)', q3_1b, a, [X],
[median, Eq(P((X >= m - a) & (X <= m + a)),
0.3)])

```

3.2 — May 2022 TZ2 Paper 1 Q8, 6 marks

A continuous random variable X has the probability density function

$$f(x) = \begin{cases} \frac{2}{(b-a)(c-a)}(x-a), & a \leq x \leq c \\ \frac{2}{(b-a)(b-c)}(b-x), & c < x \leq b \\ 0, & \text{otherwise} \end{cases}$$

— a triangle rising from 0 at a to its peak at c and back to 0 at b . Given that $c \geq \frac{a+b}{2}$, find an expression for the median of X in terms of a , b and c .

```

a, b, c, m = symbols('a b c m')

q3_2 = ...          # the median in terms of a,
b, c

X = Density({(a, c): 2*(x - a)/((b - a)*(c -
a)),
              (c, b): 2*(b - x)/((b - a)*(b -
c))}, 'X')
triangles = [{a: 0, b: 4, c: 3}, {a: 1, b: 3,
c: Rational(5, 2)}, {a: -2, b: 6, c: 5}]

verify_letters('3.2', q3_2, m, [X], [Eq(P(X <
m), 0.5)], free=triangles)

```

3.3 — November 2022 Paper 2 Q6(b), 3 marks

The continuous random variable X has a probability density function $f(x) = axe^x$ for $0 \leq x \leq b$, and 0 otherwise.

(b) In the case where $a = b = 1$, find the median of X .

3.4 — May 2024 TZ1 Paper 2 Q8(b), 4 marks

A continuous random variable X has the probability density function $f(x) = kx$ for $0 \leq x \leq k$, $f(x) = 2kx - x^2$ for $k < x \leq 2k$,

and 0 otherwise, where $k > 0$ satisfies $7k^3 = 6$.

(b) Find the median of X .

```
q3_3 = ...          # the median when a = b = 1
q3_4 = ...          # the median

m = symbols('m')
k = symbols('k', positive=True)
X = Density({(0, 1): x*exp(x)}, 'X')
Y = Density({(0, k): k*x, (k, 2*k): 2*k*x -
x**2}, 'X')

verify_letters('3.3', q3_3, m, [X], [Eq(P(X <
m), 0.5)])
verify_letters('3.4', q3_4, m, [Y], [Eq(P(Y <
m), 0.5)])
```

3.5 — May 2024 TZ2 Paper 2 Q10(c), 3 marks

The weight, in kilograms, of chocolates bought by a random customer has probability density function

$$f(x) = \begin{cases} \frac{6}{85} (4 + 3x - x^2), & 0.5 \leq x \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

(c) Find the median of X .

**3.6 — May 2025 TZ1 Paper 2 Q11(d),
3 marks**

In a marathon race, the random variable T represents the time, in hours, taken for a runner to complete the race. The probability density function for T is

$$f(t) = \begin{cases} \frac{4}{21} \left(1 - \cos \left(\frac{4\pi}{9}(t - 2.25) \right) \right), & 2.25 \leq t < 4.5 \\ \frac{4}{21} \left(1 + \cos \left(\frac{\pi}{3}(t - 4.5) \right) \right), & 4.5 \leq t \leq 7.5 \\ 0, & \text{otherwise.} \end{cases}$$

(d) Find the lower quartile of T .

```

q3_5 = ...          # the median
q3_6 = ...          # the lower quartile

m, Q1 = symbols('m Q1')
W = Density({(Rational(1, 2), 3): Rational(6,
85)*(4 + 3*x - x**2)}, 'X')
T = Density({(Rational(9, 4), Rational(9,
2)): Rational(4, 21)*(1 - cos(4*pi/9*(t -
Rational(9, 4))))),
              (Rational(9, 2), Rational(15,
2)): Rational(4, 21)*(1 + cos(pi/3*(t -
Rational(9, 2))))},
              'T', var=t)

verify_letters('3.5', q3_5, m, [W], [Eq(P(W <
m), 0.5)])
verify_letters('3.6', q3_6, Q1, [T], [Eq(P(T
< Q1), 0.25)])

```

3.7 — May 2025 TZ2 Paper 2 Q11(d), 2 marks

The time, T , in minutes that a spinning top is in motion can be modelled by the probability density function $f(t) = kte^{-3t}$ for $t \geq 0$, and 0 otherwise, where $k \in \mathbb{Z}^+$. Part (c) shows that $k = 9$.

(d) Find the median length of time that a spinning top is in motion.

Give the answer in minutes.

```
q3_7 = ...          # the median, in minutes

m = symbols('m')
top = Density({(0, oo): 9*t*exp(-3*t)}, 'T',
var=t)

verify_letters('3.7', q3_7, m, [top],
[Eq(P(top < m), 0.5)])
```

§ 4. The mode

Three questions, 10 marks. Where f is highest — and against the median.

4.1 — May 2024 TZ2 Paper 2 Q10(a), 2 marks

The weight, in kilograms, of chocolates bought by a random customer has probability density function

$$f(x) = \begin{cases} \frac{6}{85} (4 + 3x - x^2), & 0.5 \leq x \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the mode of X .

4.2 — May 2025 TZ1 Paper 2 Q11(a) (ii)-(iii), 3 marks

In a marathon race, the random variable T represents the time, in hours, taken for a runner to complete the race. The probability density function for T is

$$f(t) = \begin{cases} \frac{4}{21} \left(1 - \cos \left(\frac{4\pi}{9}(t - 2.25) \right) \right), & 2.25 \leq t < 4.5 \\ \frac{4}{21} \left(1 + \cos \left(\frac{\pi}{3}(t - 4.5) \right) \right), & 4.5 \leq t \leq 7.5 \\ 0, & \text{otherwise.} \end{cases}$$

The graph of f has a maximum point at $t = 4.5$, and $\int_{2.25}^{4.5} f(t) dt = \frac{3}{7}$.

(a) (ii) Write down the mode of T . **(iii)** Determine which is greater, the mode of T or the median of T , justifying your answer.

For (iii) answer 'mode' or 'median'.

```

q4_1 = ...          # the mode
q4_2a = ...         # the mode of T
q4_2b = ...         # 'mode' or 'median'

W = Density({(Rational(1, 2), 3): Rational(6,
85)*(4 + 3*x - x**2)}, 'X')
T = Density({(Rational(9, 4), Rational(9,
2)): Rational(4, 21)*(1 - cos(4*pi/9*(t -
Rational(9, 4)))),
              (Rational(9, 2), Rational(15,
2)): Rational(4, 21)*(1 + cos(pi/3*(t -
Rational(9, 2))))},
              'T', var=t)

verify_mode('4.1', q4_1, W)
verify_mode('4.2(a)(ii)', q4_2a, T)
verify_greater('4.2(a)(iii)', q4_2b, T)

```

4.3 — November 2025 TZ3 Paper 2 Q11(c), 5 marks

The lengths, in metres, of jumps in a long jump competition can be modelled by a continuous random variable X with probability density function

$$f(x) = \begin{cases} a(x-3)^3 + b(x-3)^2, & 3 \leq x \leq 9 \\ 0, & \text{otherwise} \end{cases}$$

where $a, b \in \mathbb{R}$, $a, b \neq 0$.

It is given that $f(9) = 0$, so that $6a + b = 0$.

(c) Show that the median of X is less than the mode of X .

Enter the mode and the median; the checks find a and b themselves.

```
a, b, m = symbols('a b m')

q4_3_mode = ...      # the mode
q4_3_median = ...    # the median

X = Density(({(3, 9): a*(x - 3)**3 + b*(x - 3)**2}, 'X')
zero = [Eq(6*a + b, 0)]

verify_mode('4.3 mode', q4_3_mode, X,
given=zero, var=[a, b])
verify_letters('4.3 median', q4_3_median, m,
[X], zero + [Eq(P(X < m), 0.5)])
```

§ 5. The mean and the variance

Six questions, 30 marks. The areas under $x f(x)$ and $x^2 f(x)$.

5.1 — May 2021 TZ2 Paper 2 Q6, 6 marks

A continuous random variable X has the probability density function $f_n(x) = (n+1)x^n$ for $0 \leq x \leq 1$, and 0 otherwise, where $n \in \mathbb{R}$, $n \geq 0$.

(a) Show that $E(X) = \frac{n+1}{n+2}$.

(b) Show that $\text{Var}(X) = \frac{n+1}{(n+2)^2(n+3)}$.

Enter the expressions in terms of n ; the checks try several n .

```
n = symbols('n', nonnegative=True)

q5_1a = ...      # E(X) in terms of n
q5_1b = ...      # Var(X) in terms of n

X = Density({(0, 1): (n + 1)*x**n}, 'X')

verify_moment('5.1(a)', q5_1a, Expect(X))
verify_moment('5.1(b)', q5_1b, Var(X))
```

5.2 — May 2022 TZ1 Paper 1 Q7(b), 4 marks

The continuous random variable X has probability density function $f(x) = \frac{k}{\sqrt{4-3x^2}}$ for $0 \leq x \leq 1$, and 0 otherwise.

(b) Find $E(X)$.

5.3 — May 2023 TZ1 Paper 2 Q11(a), 5 marks

A continuous random variable, X , has probability density function $f(x) = \frac{6}{\pi\sqrt{16-x^2}}$ for $0 \leq x \leq 2$, and 0 otherwise.

(a) Find the exact value of $E(X)$.

```
q5_2 = ...          # E(X), exact
q5_3 = ...          # E(X), exact

k = symbols('k', positive=True)
K = Density({(0, 1): k/sqrt(4 - 3*x**2)},
            'X')
X = Density({(0, 2): 6/(pi*sqrt(16 - x**2))},
            'X')

verify_moment('5.2', q5_2, Expect(K), given=
[], var=k, exact=True)
verify_moment('5.3', q5_3, Expect(X),
exact=True)
```

5.4 — May 2023 TZ2 Paper 1 Q6, 5 marks

A continuous random variable X has probability density function $f(x) = \frac{1}{2a}$ for $a \leq x \leq 3a$, and 0 otherwise, where a is a positive real number.

(a) State $E(X)$ in terms of a .

(b) Use integration to find $\text{Var}(X)$ in terms of a .

```
a = symbols('a', positive=True)

q5_4a = ...      # E(X) in terms of a
q5_4b = ...      # Var(X) in terms of a

X = Density({(a, 3*a): 1/(2*a)}, 'X')

verify_moment('5.4(a)', q5_4a, Expect(X))
verify_moment('5.4(b)', q5_4b, Var(X))
```

5.5 — May 2024 TZ2 Paper 2 Q10(e), 5 marks

The weight, in kilograms, of chocolates bought by a random customer has probability density function

$$f(x) = \begin{cases} \frac{6}{85} (4 + 3x - x^2), & 0.5 \leq x \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

The shop sells chocolates at \$25 per kilogram, but at a discounted rate of \$24 per kilogram when the weight is at least 0.75 kilograms.

(e) Find the expected amount spent per customer. Give your answer correct to the nearest cent.

5.6 — November 2025 TZ1 Paper 2
Q11(c), 5 marks

The probability density function of a random variable X is $f(x) = 3x \arccos(x^2)$ for $0 \leq x \leq k$, and 0 otherwise.

(c) Find **(i)** $E(X)$; **(ii)** $\text{Var}(X)$.

The checks find k themselves from the area being one.

```

q5_5 = ...          # the expected amount spent,
to the nearest cent
q5_6a = ...         # E(X)
q5_6b = ...         # Var(X)

W = Density({(Rational(1, 2), 3): Rational(6,
85)*(4 + 3*x - x**2)}, 'X')
spend = W.map(lambda w: Piecewise((25*w, w <
0.75), (24*w, True)), 'spend')
k = symbols('k', positive=True)
X = Density({(0, k): 3*x*acos(x**2)}, 'X')

verify_moment('5.5', q5_5, Expect(spend),
places=2)
verify_moment('5.6(c)(i)', q5_6a, Expect(X),
given=[], var=k)
verify_moment('5.6(c)(ii)', q5_6b, Var(X),
given=[], var=k)

```

§ 6. A second question around an area

Three questions, 10 marks. A ratio of areas, and an interval built from the moments.

6.1 — May 2025 TZ1 Paper 2 Q11(c), 3 marks

In a marathon race, the random variable T represents the time, in hours, taken for a runner to complete the race. The probability density function for T is

$$f(t) = \begin{cases} \frac{4}{21} \left(1 - \cos \left(\frac{4\pi}{9}(t - 2.25) \right) \right), & 2.25 \leq t < 4.5 \\ \frac{4}{21} \left(1 + \cos \left(\frac{\pi}{3}(t - 4.5) \right) \right), & 4.5 \leq t \leq 7.5 \\ 0, & \text{otherwise.} \end{cases}$$

The runners who finish the race in 3.5 hours or less are considered to be fast runners.

(c) Find the probability that a fast runner chosen at random finishes the race in 3 hours or less.

6.2 — November 2025 TZ1 Paper 2

Q11(d), 3 marks

The probability density function of a random variable X is $f(x) = 3x \arccos(x^2)$ for $0 \leq x \leq k$, and 0 otherwise.

(d) Given that $\mu = E(X)$ and $\sigma^2 = \text{Var}(X)$, find $P(\mu - \sigma < X < \mu + \sigma)$.

```
q6_1 = ...      # P(T <= 3 | T <= 3.5)
q6_2 = ...      # P(mu - sigma < X < mu +
sigma)

T = Density({(Rational(9, 4), Rational(9,
2)): Rational(4, 21)*(1 - cos(4*pi/9*(t -
Rational(9, 4))))),
            (Rational(9, 2), Rational(15,
2)): Rational(4, 21)*(1 + cos(pi/3*(t -
Rational(9, 2))))}),
            'T', var=t)
k = symbols('k', positive=True)
X = Density({(0, k): 3*x*acos(x**2)}, 'X')
mu, sigma = Expect(X), SD(X)

verify_chance('6.1', q6_1, P(T <= 3, given=T
<= 3.5))
verify_chance('6.2', q6_2, P((X > mu - sigma)
& (X < mu + sigma)), given=[], var=k)
```

6.3 — November 2025 TZ3 Paper 2

Q11(d), 4 marks

The lengths, in metres, of jumps in a long jump competition can be modelled by a continuous random variable X with probability density function

$$f(x) = \begin{cases} a(x-3)^3 + b(x-3)^2, & 3 \leq x \leq 9 \\ 0, & \text{otherwise} \end{cases}$$

where $a, b \in \mathbb{R}$, $a, b \neq 0$.

It is given that $f(9) = 0$, so that $6a + b = 0$.
Matt has the final jump in the competition. He needs to jump at least 8.52 m to win the competition.

(d) Given that he jumps over 8 m, use the model to find the probability that he wins the competition.

```
q6_3 = ...          # P(X > 8.52 | X > 8)

a, b = symbols('a b')
X = Density({(3, 9): a*(x - 3)**3 + b*(x - 3)**2}, 'X')

verify_chance('6.3', q6_3, P(X > 8.52,
given=X > 8), given=[Eq(6*a + b, 0)], var=[a,
b])
```



Solutions

$$1.1 \int_0^{0.5} f(x) \, dx = 0.239358 \dots = \boxed{0.239}.$$

$$1.2 \int_1^2 f(x) \, dx = \frac{37}{85} = 0.435294 \dots = \boxed{0.435}.$$

$$1.3 \int_{2.25}^{3.5} f(t) \, dt = 0.103749 \dots = \boxed{0.104}.$$

1.4 Below 0.75 kg nobody spends more than \$18.75; from 0.75 kg, $24x \leq 48$ gives $x \leq 2$. $\int_{0.5}^2 f(x) \, dx = 0.635294 \dots = \boxed{0.635}$.

1.5 Above 1.5 for 0.337978... s in each period of 0.805536... s, six full periods in five seconds: $\frac{6 \times 0.337978 \dots}{5} = \boxed{0.406}$.

$$2.1 \text{ (a) } \int_0^4 \frac{x}{(x^2+k)^{3/2}} \, dx = \boxed{\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{16+k}}} = 1,$$

and multiplying by $\sqrt{k}\sqrt{16+k}$ gives the result. (b) $k = 0.645038 \dots = \boxed{0.645}$.

$$2.2 \frac{k}{\sqrt{3}} \cdot \frac{\pi}{3} = 1: \boxed{k = \frac{3\sqrt{3}}{\pi}}.$$

$$2.3 a[xe^x - e^x]_0^b = 1: \boxed{a = \frac{1}{be^b - e^b + 1}}.$$

$$\mathbf{2.4} \quad \frac{k^3}{2} + (3k^3 - \frac{7}{3}k^3) = \boxed{\frac{7k^3}{6}} = 1, \text{ so } 7k^3 = 6.$$

$$\mathbf{2.5} \quad (\mathbf{a}) \quad \left[\frac{a}{4}(x-3)^4 + \frac{b}{3}(x-3)^3 \right]_3^9 = \frac{324a + 72b}{216a + 36b} = 1. \quad (\mathbf{b})(\mathbf{i}) \quad f(9) = 216a + 36b = 0, \text{ and dividing by } 36, 6a + b = 0.$$

$$\mathbf{3.1} \quad (\mathbf{a}) \quad m \arccos m - \sqrt{1 - m^2} + 1 = 0.5: m = 0.360034 \dots = \boxed{0.360}. \quad (\mathbf{b}) \quad \int_{m-a}^{m+a} \arccos x \, dx = 0.3: a = 0.124861 \dots = \boxed{0.125}.$$

$$\mathbf{3.2} \quad \text{The triangle up to } m \text{ has area } \frac{(m-a)^2}{(b-a)(c-a)} = \frac{1}{2}, \quad \text{and} \quad m \geq a:$$

$$\boxed{m = a + \sqrt{\frac{(b-a)(c-a)}{2}}}.$$

$$\mathbf{3.3} \quad me^m - e^m + 1 = \frac{1}{2}: m = 0.768039 \dots = \boxed{0.768}.$$

$$\mathbf{3.4} \quad k = 0.949914 \dots; \text{ the first piece holds } \frac{3}{7}, \text{ so the median is on the second: } m = 1.02925 \dots = \boxed{1.03}.$$

$$\mathbf{3.5} \quad \int_{0.5}^m f(x) \, dx = 0.5: m = 1.68701 \dots = \boxed{1.69} \text{ kg.}$$

3.6 The first piece holds $\frac{3}{7} > \frac{1}{4}$: $q = 4.01290\dots = \boxed{4.01}$.

3.7 $1 - (3m + 1)e^{-3m} = 0.5$: $m = 0.559448\dots = \boxed{0.559}$ minutes (33.6 seconds).

4.1 The parabola $4 + 3x - x^2$ has its vertex at $x = 1.5$: mode $\boxed{1.5}$ kg.

4.2 (ii) $\boxed{4.5}$. (iii) $P(T < 4.5) = \frac{3}{7} < 0.5$, so the median (4.69) is greater: $\boxed{\text{median}}$.

4.3 $a = -\frac{1}{108}$, $b = \frac{1}{18}$. The maximum of f is at $(7, 0.296296\dots)$: mode $\boxed{7}$. The area from 3 to 7 is $0.592592\dots > 0.5$; the median is $6.68563\dots = \boxed{6.69} < 7$.

5.1 (a) $\int_0^1 (n+1)x^{n+1} dx = \boxed{\frac{n+1}{n+2}}$. (b)

$E(X^2) = \frac{n+1}{n+3}$, and $\frac{n+1}{n+3} - \left(\frac{n+1}{n+2}\right)^2 = \boxed{\frac{n+1}{(n+2)^2(n+3)}}$.

5.2 $k = \frac{3\sqrt{3}}{\pi}$, and $\frac{3\sqrt{3}}{\pi} \left[-\frac{1}{3}\sqrt{4-3x^2} \right]_0^1 = \boxed{\frac{\sqrt{3}}{\pi}}$.

$$\mathbf{5.3} \quad \frac{6}{\pi} \left[-\sqrt{16-x^2} \right]_0^2 = \boxed{\frac{12}{\pi} (2 - \sqrt{3})}.$$

$$\mathbf{5.4} \quad (\text{a}) \quad \boxed{2a} \text{ by symmetry. } (\text{b}) \quad \frac{13a^2}{3} - 4a^2 = \boxed{\frac{a^2}{3}}.$$

$$\mathbf{5.5} \quad \int_{0.5}^{0.75} 25x f(x) dx + \int_{0.75}^3 24x f(x) dx = 40.9576 \dots: \boxed{\$40.96}.$$

$$\mathbf{5.6} \quad k = 0.713250. \quad (\text{i}) \quad 0.456309 \dots = \boxed{0.456}. \quad (\text{ii}) \quad 0.237198 \dots - 0.456309 \dots^2 = 0.0289805 \dots = \boxed{0.0290}.$$

$$\mathbf{6.1} \quad \frac{P(T \leq 3)}{P(T \leq 3.5)} = \frac{0.0247152 \dots}{0.103749 \dots} = 0.238220 \dots = \boxed{0.238}.$$

$$\mathbf{6.2} \quad \sigma = 0.170236 \dots; \quad \int_{0.286072 \dots}^{0.626545 \dots} f(x) dx = 0.620012 \dots = \boxed{0.620}.$$

$$\mathbf{6.3} \quad \frac{P(X > 8.52)}{P(X > 8)} = \frac{0.0344268 \dots}{0.131944 \dots} = 0.260919 \dots = \boxed{0.261}.$$